SHANGYANG MIN

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EDUCATION

Brown University — Sep 2023 – May 2025
M.Sc. in Computer Science | GPA: 4.0/4.0 | Multimodal AI, Brain-Inspired Computing

Michigan State University — Aug 2019 – May 2023
B.S. in Computer Science Engineering, High Honor | GPA: 3.95/4.0 | ML, Computer Vision, Biomedical Imaging

EXPERIENCE

Hubei Lianhuashan AI Research Institute — Jun 2025 – Present
Research Engineer | Under Peking University Institute for General AI | Ezhou, China

- Architected an AI Agent framework for traditional industries, enabling non-technical enterprises to deploy domain-specific agents through standardized tool integration and deterministic workflow orchestration.
- Led multi-track AI education programs—executive workshops, technical bootcamps, and university collaborations with Hubei institutions—training participants across enterprise and academic audiences.

Abysen AI — Feb 2025 – Present
Founder | AI Research Startup | United States | abysen.com

- Founded a startup building personalized AI Twin systems including: digital persona replication, behavior cloning and memory, and RelOS (voice-driven relationship memory compiler).

Lee Lab, Brown University — Sep 2024 – May 2025
Research Assistant | Brain-Computer Interface & Multimodal Interaction

- Built a real-time EEG-to-Unity pipeline for brain-signal avatar control and developed a multimodal fusion architecture (vision + EEG) improving cross-modal interaction fidelity.

HAAI Lab, Michigan State University — May 2022 – Aug 2023
Research Assistant | Biomedical Imaging | Funded by NSF & HFH+MSU Cancer Grant

- Proposed Feature Imitating Networks (FINs) that distill large-model representations into lightweight CNNs, cutting inference time by 3× while boosting accuracy—published at IEEE EMBC 2024 with open-source release.

Henry Ford Health × Michigan State University — Sep 2022 – Sep 2023
Research Assistant | NIH-Funded Program | Tumor Detection & Radiomics

- Built ML pipelines for tumor detection from DCE-MRI with radiomics feature extraction; developed standardized preprocessing for multi-institutional cross-site model evaluation.

SELECTED PROJECTS

LLM-Agent-Core: Open-Source LLM & Agent Curriculum

- Designed and open-sourced a 12-chapter hands-on course (Autograd → Transformer → SFT/DPO → Agent) with runnable notebooks; includes ReAct, RAG, Code Agent, and Multi-Agent implementations with a unified LLM backend abstraction.

Trigger-Simulation Backdoor Removal Defense (TSBRD)

- Proposed virtual-prompt realignment for backdoored LLMs, reducing attack success rate from 93% → 0% on Llama-2 within 7 epochs while preserving MMLU accuracy.

VQA with Action Insights

- Integrated SOTA vision models as tokenized video-info annotations into LLM prompts, matching fine-tuned SeViLa with zero-shot GPT-4 inference.

scCST: Continuous Spatial-Temporal Transformer

- Developed a transformer for interpolating cellular trajectories in single-cell sequencing data, enabling continuous modeling of gene expression dynamics.

Additional research projects: shangyangmin.com/projects

TECHNICAL SKILLS

Core: Python, PyTorch, Hugging Face (Transformers/PEFT/TRL), C/C++, SQL
Infra & Tools: CUDA, vLLM, Ollama, Docker, Git, W&B, Linux, LaTeX
Interests: LLM/Agent Systems, Multimodal Learning, Computer Vision, Diffusion Models, BCI

PUBLICATION

S. Min, H. B. Ebadian, T. Alhanai, M. M. Ghassemi, *“Feature Imitating Networks Enhance the Performance, Reliability and Speed of Deep Learning on Biomedical Image Processing Tasks,”* IEEE EMBC 2024.